

Neuron

Mathematical Foundations of a Self-Organizing Cognitive Memory Graph

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Abstract

Persistent, self-organizing memory is essential for autonomous AI agents to achieve long-term coherence. Traditional RAG systems often suffer from factual fragmentation and static retrieval. This paper introduces **Neuron**, a graph-based cognitive memory framework where beliefs (nodes) and associations (edges) evolve dynamically. We detail Neuron’s multi-stage architecture—comprising surprise filtering, semantic abstraction, adaptive retrieval evolution, and a biologically-inspired “Sleep Cycle” for memory consolidation. We provide the mathematical foundations for novelty scoring, confidence reinforcement, and strategy optimization, offering a robust substrate for lifelong agent learning.

1. Introduction

Modern Large Language Models (LLMs) possess vast parametric knowledge but lack the ability to autonomously update their worldview based on new experiences without expensive fine-tuning. Traditional memory systems are typically implemented as flat vector databases or short-term retrieval pipelines, resulting in:

- Fragmented factual storage
- Weak contextual reasoning
- Redundant semantic accumulation
- Poor long-term coherence

Neuron addresses these limitations through a persistent graph-backed cognitive memory architecture inspired by biological learning systems. Instead of treating memory as isolated embeddings, Neuron organizes knowledge into a continuously evolving semantic graph where:

- **Nodes** represent distilled semantic beliefs
- **Edges** represent contextual associations
- **Retrieval** reinforces semantic pathways
- **Maintenance cycles** reorganize memory

The framework transforms memory from passive storage into a self-organizing cognitive substrate capable of:

1. Incremental learning
2. Semantic abstraction
3. Adaptive retrieval optimization

4. Memory consolidation
5. Long-term contextual reasoning

By integrating vector similarity search with graph traversal dynamics and reinforcement-based adaptation, Neuron provides a foundation for persistent autonomous cognition.

2. Data Representation

The fundamental unit of knowledge in Neuron is the **Belief**, represented as a graph node. Semantic and contextual relationships are encoded as weighted graph edges.

2.1 Node Structure (Belief)

Each node n is a structured semantic object containing the following attributes:

Table 1: Belief Node Structure

Field	Description
ID	Unique UUID identifier
User ID	Multi-tenant ownership identifier
Label	Distilled semantic fact
Confidence (C)	Certainty score in $[0, 1]$
Evidence Count	Number of confirmations or observations
Contradiction Count	Frequency of semantic conflicts identified
Domain Tags	Semantic categories (e.g. AI, Finance)
Entities	Extracted named entities
Temporal Stability	{stable, volatile, time-bound}
Abstraction Level	{specific, pattern, principle}
Deprecated	Boolean flag for resolution/forgetting
Embedding	384-dimensional semantic vector
Created At	Timestamp of node formation
Last Confirmed At	Timestamp of most recent evidence reinforcement
Last Contradicted At	Timestamp of most recent conflict identified

The semantic embedding is represented as:

$$E(n) \in \mathbb{R}^{384}$$

using the default encoder `all-MiniLM-L6-v2`.

2.2 Edge Structure (Association)

Edges encode contextual relationships between beliefs. Each edge e_{ij} contains:

3. Scoring Taxonomy and Input Mechanics

Neuron’s operation is governed by several adaptive scoring mechanisms.

Table 2: Association Edge Structure

Field	Description
ID	Unique edge UUID identifier
From / To ID	UUIDs of connected nodes
Relation Type	supports , contradicts , refines , depends_on , ...
Weight (w)	Semantic strength in $[0.1, 1.0]$
Evidence Count	Number of confirmations or traversals
Created At	Timestamp of edge formation
Last Updated At	Timestamp of most recent weight reinforcement

Table 3: Neuron Scoring Metrics

Metric	Formula / Logic	Range / Threshold
Novelty (N)	$1 - \max(\text{cosine_sim}(V_{in}, V_{neighbors}))$	0.0–1.0; novel if $N > 0.3$
Confidence (C)	$C_{t+1} = \min(0.95, C_t \times 1.1)$ per confirmation	0.0–1.0; valid if $C > 0.35$
Edge Weight (w)	$w_{t+1} = \min(1.0, w_t \times 1.05)$ per traversal	0.1–1.0; active if $w > 0.6$
Fitness (F)	EMA: $(1 - \alpha)F_t + \alpha R$, $\alpha = 0.3$	0.0–1.0
Implicit Score (S_i)	$0.5 + 0.2(h > 0) + 0.1(h > 5)$	0.5–0.8

3.1 Novelty Score

Novelty quantifies semantic surprise relative to existing memory:

$$N = 1 - \max_{b \in B} (\text{cosine_sim}(E(T), E(b)))$$

where T is the incoming text, $E(\cdot)$ is the embedding function, and B is the current belief set.

3.2 Confidence Reinforcement

Confidence increases multiplicatively with repeated confirmations:

$$C_{t+1} = \min(0.95, C_t \times 1.1)$$

This approach provides larger relative gains for low-confidence beliefs while decelerating as they approach the 0.95 saturation limit.

3.3 Fitness Score

Retrieval strategies are evaluated using an Exponential Moving Average (EMA):

$$F_{t+1} = 0.7 F_t + 0.3 R$$

where F_t is the prior strategy fitness and R is the retrieval reward derived from:

1. Explicit user feedback
2. Implicit retrieval heuristics (see Section 3.4)

3.4 Implicit Score

When explicit feedback is absent, retrieval quality is estimated via the implicit score S_i :

$$S_i = 0.5 + 0.2 (h > 0) + 0.1 (h > 5)$$

where h denotes the number of relevant belief nodes (hits) successfully retrieved. This mechanism is heavily used during maintenance cycles to evolve strategies in the background.

4. Ingestion and Surprise Filtering

Neuron uses a staged ingestion pipeline designed to minimize redundancy while preserving semantically valuable information.

4.1 Surprise Filter Gate

For every incoming text chunk T , the system computes:

$$S_{\max} = \max_{b \in \text{NN}(T, 5)} (\text{cosine_sim}(E(T), E(b)))$$

where $\text{NN}(T, 5)$ denotes the five nearest semantic neighbors.

The ingestion pipeline is divided into three semantic bands.

4.1.1 Novelty Band

Condition: $1 - S_{\max} \geq 0.3$

- Input is considered semantically novel
- Sent to the LLM abstraction pipeline
- A new belief node is created

4.1.2 Confirmation Band

Condition: $0.75 \leq S_{\max} \leq 0.90$

- Existing node evidence count incremented
- Confidence reinforced

4.1.3 Redundancy Band

Condition: $S_{\max} > 0.90$

- Input discarded

- Prevents semantic duplication
- Reduces memory bloat

5. Data Addition Methods

5.1 Single-Chunk Pipeline

The primary real-time ingestion pipeline consists of:

1. **Surprise Check.** If $N < 0.3$, the system updates or skips existing beliefs.
2. **LLM Abstraction.** Novel content is distilled into:
 - Semantic labels
 - Domain tags
 - Entities
 - Abstraction level
 - Temporal stability
3. **Node Creation.** A new belief node is inserted into the graph.
4. **Local Linking.** Edges are created to the two nearest neighbors, with w = similarity.

5.2 Fast Batch Mode

Designed for high-throughput ingestion, the system first computes the self-similarity matrix:

$$M = VV^\top$$

Any chunk i satisfying $\exists j < i : M_{ij} > 0.95$ is discarded. Remaining chunks are connected through k -NN graph construction ($k = 3$), with edge weights initialized using cosine similarity.

5.3 Deep Batch Mode

Deep Batch mode prioritizes semantic abstraction and long-range linkage:

1. **Windowed LLM Abstraction.** Default window size: $n = 20$.
2. **Global Entity Linking.** For each extracted entity:
 1. Search the graph for the strongest matching entity node
 2. Select the node with highest evidence count
 3. Create a cross-batch semantic edge with $w = 0.9$

This creates persistent semantic hubs across datasets and time.

6. Retrieval and Myelination

Neuron retrieval combines vector similarity with graph traversal.

6.1 Graph Traversal

Retrieval proceeds in two stages:

1. Vector seed selection ($k = 5$ seeds)
2. Breadth-First graph traversal to depth $D = 3$

Edges satisfying $w < 0.6$ are treated as high-resistance pathways and ignored.

6.2 Myelination

Traversed edges are reinforced after successful retrieval:

$$w_{t+1} = \min(1.0, w_t \times 1.05)$$

This creates emergent semantic routing optimization analogous to biological myelination.

7. Adaptive Strategy Evolution

Neuron autonomously evolves retrieval strategies.

7.1 Selection and Exploration

The framework uses an ϵ -greedy policy with $\epsilon = 0.2$:

- **Explore:** random strategy selected with probability ϵ
- **Exploit:** best-fitness strategy selected with probability $1 - \epsilon$

7.2 Evolution Cycle

Evolution is triggered every 50 retrieval events:

1. Rank all strategies by fitness F
2. Remove the lowest-fitness strategy
3. Clone the best-performing strategy
4. Apply stochastic mutation (see below)

7.3 Mutation Functions

Child parameters are derived from the parent via:

$$k_{\text{child}} = \text{round}(k_{\text{parent}} + \mathcal{U}(-1, 1)) \quad (1)$$

$$D_{\text{child}} = \text{round}(D_{\text{parent}} + \mathcal{U}(-1, 1)) \quad (2)$$

$$w_{\text{min,child}} = w_{\text{min,parent}} + \mathcal{N}(0, 0.1) \quad (3)$$

$$\tau_{\text{child}} = \tau_{\text{parent}} + \mathcal{N}(0, 0.1) \quad (4)$$

where k is the retrieval seed count, D the traversal depth, w_{min} the minimum traversable edge weight, and τ the similarity threshold.

8. Maintenance: The Sleep Cycle

Neuron periodically enters a maintenance phase inspired by biological sleep consolidation.

8.1 Definitions

Maintenance — Low-level graph operations:

- Pruning
- Consolidation
- Coherence repair

Sleep Cycle — High-level orchestration phase responsible for:

1. Implicit scoring
2. Strategy evolution
3. Graph maintenance

Maintenance cycles execute every 24 hours.

8.2 Internal Workflow

8.2.1 1. *Implicit Scoring*

Unscored retrieval events are evaluated using:

$$S_i = 0.5 + 0.2 (\text{hits} > 0) + 0.1 (\text{hits} > 5)$$

The resulting scores update retrieval strategy fitness.

8.2.2 2. *Pruning*

Beliefs satisfying both of the following conditions are deprecated or removed:

- No confirmation within the past 30 days
- $C < 0.35$

This minimizes graph entropy and preserves semantic efficiency.

8.2.3 3. *Consolidation*

When ≥ 5 nodes share a common entity hub, the system synthesizes a higher-level abstraction called a **Master Belief**. The Master Belief inherits:

- The maximum confidence among constituents
- The sum of constituent evidence counts

This enables the organic emergence of generalized semantic concepts.

9. Emergent Cognitive Behavior

Through repeated retrieval, reinforcement, pruning, and consolidation, Neuron exhibits several emergent properties:

1. **Semantic Path Optimization.** Frequently traversed reasoning pathways become dominant.
2. **Contextual Memory Clustering.** Related beliefs self-organize into semantic hubs.
3. **Long-Term Concept Formation.** Repeated observations evolve into generalized princi-

ples.

4. **Adaptive Retrieval Evolution.** Retrieval policies improve autonomously over time.
5. **Dynamic Knowledge Compression.** Consolidation reduces redundancy while preserving semantic richness.

10. Conclusion

Neuron transforms memory from passive storage into a continuously evolving cognitive substrate. By integrating:

- Vector similarity search
- Graph-structured semantic relationships
- Reinforcement-driven pathway optimization
- Adaptive retrieval evolution
- Biologically-inspired consolidation mechanisms

the framework enables persistent self-organizing agent intelligence. Unlike traditional RAG systems that retrieve isolated fragments, Neuron maintains an evolving semantic worldview in which beliefs, associations, confidence, and abstraction continuously co-adapt over time.

The architecture provides a foundation for future autonomous systems capable of:

1. Lifelong learning
2. Dynamic memory consolidation
3. Contextual reasoning
4. Adaptive strategy evolution
5. Coherent long-term cognition

As AI agents increasingly operate across extended time horizons and complex environments, systems such as Neuron may become foundational infrastructure for scalable machine cognition.

— *End of Paper* —